Solution Approach & Design

1. Overall Approach

The proposed solution is a digital system that replaces manual processes with automated data entry, validation, and processing. It focuses on improving accuracy, transparency, and efficiency while maintaining a simple and scalable design.

2. System Architecture Overview

The system follows a **layered architecture**:

- **Frontend Layer:** User interface for staff and farmers
- **Backend Layer:** Handles business logic and APIs
- **Database Layer:** Stores all data securely

This separation ensures scalability and maintainability.

3. Key Components

- **Farmer Management Module:** Handles registration and data
- **Milk Collection Module:** Records daily milk entries
- **Quality Analysis Module:** Evaluates milk quality
- **Billing Module:** Calculates payments automatically
- **Reporting Module:** Generates reports and summaries

4. Data Flow Explanation

1. User enters milk data through frontend
2. Data is sent to backend via API
3. Backend validates input
4. Services process quality and payment
5. Data is stored in database

6. Results are displayed on dashboard and reports

5. Design Decisions

- **Use of MongoDB:** For flexible and scalable data storage
- **Service-Based Architecture:** To separate business logic
- **JWT Authentication:** For secure access control
- **Modular Design:** For easy extension and maintenance
- **Focus on MVP:** To prioritize core functionality